Fuel Queue Management System

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1. Introduction

Sri Lanka has experienced severe fuel shortages in recent years. During these periods, people often spend several hours waiting in long queues at fuel stations without knowing whether fuel is actually available. This situation leads to wasted time, traffic congestion, frustration among citizens, and inefficient fuel distribution.

To address this issue, we propose a **Fuel Queue Management System**, a web-based platform that allows users to check real-time fuel availability and queue conditions at nearby fuel stations. The system will also allow users and station administrators to update queue status and fuel availability, helping citizens make informed decisions before traveling to a station.

The main goal of this project is to **reduce unnecessary waiting time and improve fuel distribution awareness** using web technologies.

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2. Problem Identification

2.1 Problem Description

Fuel shortages in Sri Lanka cause:

- Long waiting queues at fuel stations
- Lack of information about fuel availability
- Traffic congestion around stations
- Time and fuel wastage for citizens searching for fuel

Currently, there is **no centralized platform** where people can check real-time fuel queue status or availability before visiting a fuel station.

2.2 Impact of the Problem

The problem affects several groups:

General Public

- Spend hours waiting in queues
- Travel between multiple stations unnecessarily

Fuel Station Operators

- Difficulty managing crowd and queue flow

Government / Authorities

- Traffic congestion and public dissatisfaction

A digital solution can significantly reduce these issues by providing **real-time queue and availability updates**.

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3. Proposed Solution

3.1 System Overview

The proposed solution is a **Fuel Queue Management Website** that allows users to:

- Check nearby fuel stations
- View real-time queue status
- Check fuel availability
- Report queue updates
- Receive notifications when fuel becomes available

The system will allow **both users and station administrators** to update fuel and queue information.

3.2 Key Features

User Features

- User registration and login
- Search fuel stations by location
- View queue length and waiting time
- View fuel availability (Petrol/Diesel)
- Submit queue reports with optional photos
- Receive alerts when fuel becomes available

Admin / Station Features

- Update fuel availability
- Update queue status
- Manage reports
- Monitor user submissions

3.3 User Journey

Typical user interaction flow:

- User visits the website
- User registers or logs in
- User enters location or selects area
- System displays nearby fuel stations
- User views queue status and fuel availability
- User decides which station to visit
- Users can submit reports about queue length
- Station admins update official fuel status

This process allows the community to share real-time information.

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4. Technical Specifications

4.1 Front-End Technologies

The front-end will be designed to provide a responsive and user-friendly interface.

Technologies used:

HTML5 – Website structure

CSS3 – Styling and layout

Bootstrap – Responsive design

JavaScript – Interactive features

Google Maps API – Display fuel station locations

Design considerations:

- Mobile-friendly interface
- Simple navigation
- Clear queue and fuel status indicators

4.2 Back-End Development

The server-side system will be developed using **PHP**.

Functions of the backend:

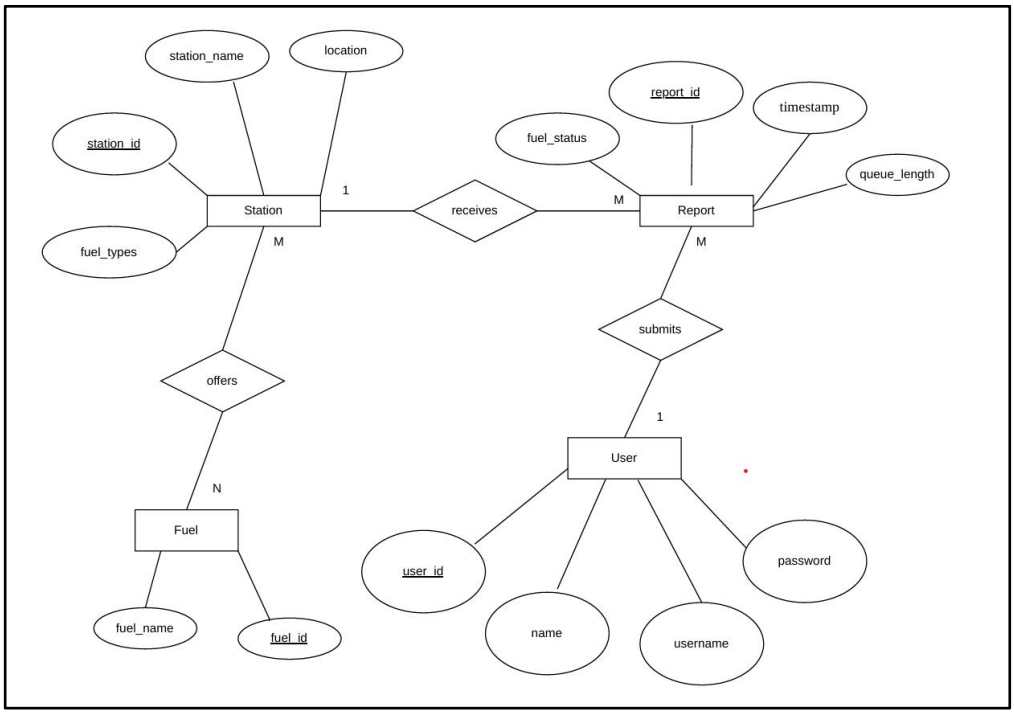
- User authentication (login/register)
- Handling fuel station data
- Managing queue reports
- Updating fuel availability
- Processing user submissions
- Providing API endpoints for the frontend

PHP will communicate with the database to store and retrieve data efficiently.

4.3 Database Design

The system will use **MySQL** for database management.

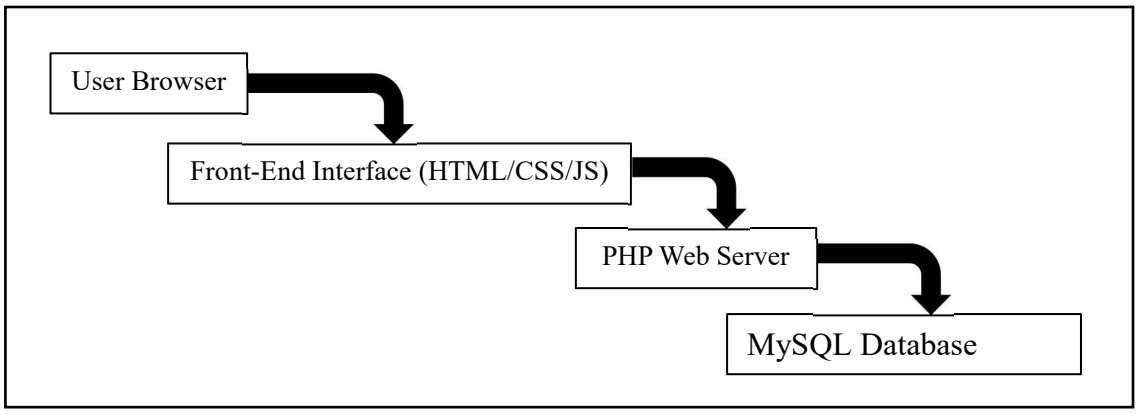
ER Diagram:



This structure allows efficient storage and retrieval of fuel and queue information.

4.4 System Architecture

System architecture flow:



The user interacts with the interface, which sends requests to the PHP server. The server processes the request and retrieves or stores data in the MySQL database.

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5. Development Plan - Gantt chart

The project will be developed over 6 weeks with the following milestones.

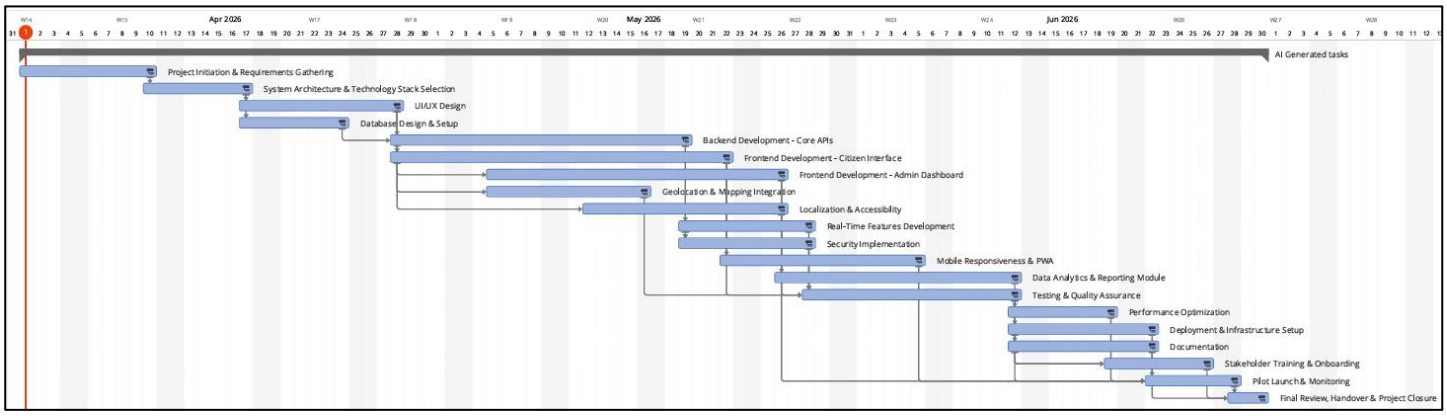
Week 1: Research and requirement analysis, System planning, Database design

Week 2 – Week 3: Front-end interface development, User registration and login system, Station search functionality

Week 4: Database integration, Report submission feature, Fuel status updates

Week 5: Notification features, Admin dashboard development, Testing and debugging

Week 6: Final integration, System testing, Deployment and demonstration



6. Group Contributions

If the project is done as a pair. Both of us act as:

Front-End Developer: UI/UX design, Responsive interface, Google Maps integration, User interaction features

Back-End Developer: PHP back-end development, User authentication, API development, System logic implementation

Database & Testing: MySQL database design, Data integration, System testing, Bug fixing and documentation

Each member will contribute equally to ensure successful project completion.

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7. Budget Estimation

Estimated costs for the project are relatively low.

Item	Estimated Cost
Domain (.lk)	$10 per year
Web Hosting	$20 - $50 per year
Development Tools	Free (XAMPP, VS Code, Bootstrap)

Total estimated cost: **Under $100**

Annual maintenance cost: approximately **$20 – $30**.

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8. Conclusion

The Fuel Queue Management System aims to solve a significant real-world problem faced by Sri Lankan citizens during fuel shortages. By providing real-time queue information and fuel availability updates, the system helps users make better decisions and reduces unnecessary waiting time.

Using modern web technologies such as **PHP, MySQL, HTML, CSS, and JavaScript**, the project can be developed within the semester timeline. The platform has the potential to improve fuel station efficiency and enhance public convenience.

This project demonstrates how web technology can be used to create practical solutions for real-world challenges.